INCH-POUND
MIL-DTL-32687/2
w/Amendment 2
18 July 2025
SUPERSEDING
MIL-DTL-32687/2
w/Amendment 1
15 May 2023

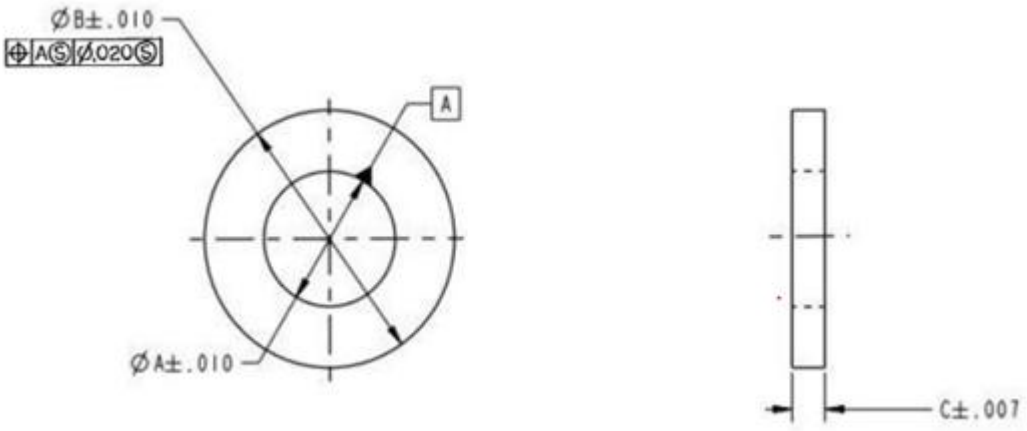
DETAIL SPECIFICATION SHEET

WASHER, FLAT, ALLOY OR CARBON STEEL, ROUND, HARDENED, ZINC-NICKEL PLATED, EXTRA LARGE

(Similar to 19207-10941915)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32687.



NOTES:
1. Dimensions are in inches.
2. If stamping is used to manufacture washers, tolerance for dimensions A and B shall be + 0.040/ -0.010.
3. All edges and corners shall be free from burrs.
4. Unless otherwise specified, edges shall be broken R 0.02+ 0.02/ - 0.00.

FIGURE 1. Washer, flat, alloy or carbon steel, hardened, extra large.

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AMSC N/A FSC 5310

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REQUIREMENTS:

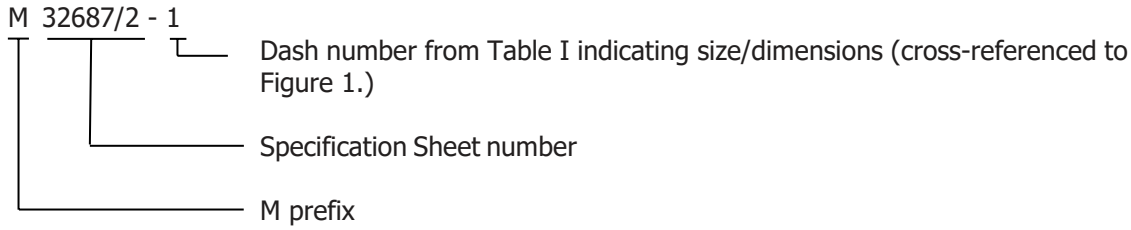
1 Steel composition. Unless otherwise specified, washers shall be made from one of the following materials:

Steel, alloy, strip, sheet or plate, grade or designation 4130, 4140, or 8640, spec ASTM A506, or ASTM A829/A829M
Steel, carbon, strip, designation 1050, spec ASTM A684/A684M
Steel, alloy, bar, grade 4130, 4140, 8640, or 8740, spec ASTM A108 or ASTM A322
Steel, alloy, strip, sheet or plate, 4130, spec SAE AMS6350, SAE AMS6351 or SAE AMS6345

2 Hardness. Washers shall be quenched and tempered to Rockwell C33 through 38 (or equivalent hardness scale).

3 Finish. Zinc-nickel plating per MIL-PRF-32647, Type K20, or zinc-nickel plating per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS.

4 Part number. Part or Identifying Number (PIN) consists of the following:



Example: M32687/2-1 is a Flat Washer – Nominal Size 0.250, ID 0.281, OD 0.750, and 0.090 inch thickness with zinc-nickel, Type K20 Plating per MIL-PRF-32647, or Zinc-Nickel per ASTM F1941/F1941M, Fe/Zn-Ni 8BNS. (See Table II for cross-reference from part number 19207-10941915 to PIN).

TABLE I. Dimensions.

PIN	Nominal Size	A	B	C
M32687/2-1	0.250	0.281	0.750	0.090
M32687/2-2	0.313	0.343	0.875	0.090
M32687/2-3	0.375	0.406	1.000	0.104
M32687/2-4	0.438	0.468	1.250	0.119
M32687/2-5	0.500	0.531	1.375	0.134
M32687/2-6	0.563	0.594	1.500	0.164
M32687/2-7	0.625	0.656	1.750	0.179
M32687/2-8	0.750	0.812	2.000	0.194
M32687/2-9	0.875	0.938	2.250	0.250
M32687/2-10	1.000	1.062	2.500	0.280
M32687/2-11	1.125	1.250	2.750	0.320
M32687/2-12	1.250	1.375	3.000	0.360
M32687/2-13	1.375	1.500	3.250	0.400
M32687/2-14	1.500	1.625	3.500	0.450

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TABLE II. Old and new PIN. /1, /2 /3

Old Part Number	New Part Number
10941915-1	M32687/2-1
10941915-2	M32687/2-2
10941915-3	M32687/2-3
10941915-4	M32687/2-4
10941915-5	M32687/2-5
10941915-6	M32687/2-6
10941915-7	M32687/2-7
10941915-8	M32687/2-8
10941915-9	M32687/2-9
10941915-10	M32687/2-10
10941915-11	M32687/2-11
10941915-12	M32687/2-12
10941915-13	M32687/2-13
10941915-14	M32687/2-14

/1 The difference between the parts from the Old and New parts is the finish.
/2 Old part numbers are not to be used for new design.
/3 Use new part numbers in place of old part numbers with design authority approval.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 70

Preparing activity:
Army – AT

(Project 5310-2025-021)

Review activities:
Army – AR, AV, CR4, MI
Navy – OS, SH
Air Force – 11, 20

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